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Flat rack with extender

Abstract

An adjustable rack system, a shelving system, and methods are provided. The adjustable rack system includes an adjustable frame and an adjustable rack. The adjustable frame includes a front rail, a rear rail and at least one longitudinally adjustable rail. The longitudinally adjustable rail interconnects the front and rear rails. The longitudinally adjustable rail is longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis. The adjustable rack is removably attached to the adjustable frame. The adjustable rack includes a first panel defining a rear edge of the adjustable rack and a second panel defining a front edge of the adjustable rack. The second panel is adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack. The adjustable rack defines a top surface defined in part by each of the first and second panels.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This patent application claims the benefit of U.S. Provisional Patent Application No. 63/253,832, filed Oct. 8, 2021, the entire teachings and disclosure of which are incorporated herein by reference thereto.

FIELD OF THE INVENTION

(1) This invention generally relates to retail display systems and adjustable racks for retail display systems.

BACKGROUND OF THE INVENTION

(2) Retail display systems such as shelving systems are used for displaying merchandise. One particularly shelving system is a refrigerated case that has one or more shelves for displaying merchandise. Some refrigerated cases often have a well formed at a bottom region thereof. The bottom most shelf, which may also be referred to as a rack, often covers a portion of the well that holds components of the system such as pumps, controllers, electrical wiring, hoses, refrigeration components, etc.

(3) Overtime, dust, dirt, grime, and product can get into the well and below the bottom most shelf. Further, maintenance is often required on the components of the system stored below the bottom most shelf. Easy and efficient access to the region below the bottom most shelf can make cleaning easier and more cost effective.

(4) Often, the bottom most shelf is provided by a wire shelf made from interconnected pieces of round wire. Typically, the surface area of these shelves have a greater percentage of openings than surface area provided by the wires. This can make these shelves difficult to clean. For example, it is quite often required to clean around the wire rather than simply clean a flat surface, making complete cleaning difficult.

(5) Further, by being made from wire, these shelves can be relatively heavy.

(6) Further, these wire shelves typically require entire removal of the shelf from the display system to access the well below the shelves. One problem when an entire shelf is removed, it may be difficult to remember the proper orientation to reinstall the shelves. This is particularly true if the person cleaning the shelf is not the person who originally staged the shelving system or that has to reassemble the shelving system after cleaning. For instance, it is quite often that the person cleaning the shelving system is not the person who reassembles the shelving system as the shelving system may need a period of time to dry prior to reassembly.

(7) In some systems, there are plenums below the bottom most product shelf/rack that cover components of the system as noted above (e.g. pumps, fans, hoses, controllers, etc.). In prior systems, it was often necessary to retire the entire bottom most shelf to provide access to or movement of the plenums to allow for cleaning thereunder.

(8) Examples of the present application provide improvements over the current state of the art for shelving systems and particularly shelving systems that have a well formed at the bottom region thereof.

BRIEF SUMMARY OF THE INVENTION

(9) In an example, an adjustable rack system for mounting to a frame assembly of a shelving unit is provided. The adjustable rack system includes an adjustable frame and an adjustable rack. The adjustable frame includes a front rail, a rear rail and at least one longitudinally adjustable rail. The rear rail is spaced from the front rail. The longitudinally adjustable rail interconnects the front and rear rails. The at least one longitudinally adjustable rail is longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis. The adjustable rack is

removably attached to the adjustable frame. The adjustable rack includes a first panel defining a rear edge of the adjustable rack and a second panel defining a front edge of the adjustable rack. The second panel is adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack. The adjustable rack defines a top surface defined in part by each of the first and second panels.

(10) In one example, the at least one longitudinally adjustable rail includes a first and a second side rail that interconnect the rear and front rails. The second side rail is laterally spaced apart from the first side rail. The first and second side rails being longitudinally adjustable to adjust a length thereof parallel to the adjustment axis.

(11) In one example, each of the first and second side rails includes a front side rail member and a rear side rail member. The front side rail member is adjustably positionable relative to the rear side rail member to adjust the length of the corresponding side rail. The front side rail members are attached to the front rail. The rear side rail members are attached to the rear rail.

(12) In one example, the first panel is operably attached to the rear rail and the second panel is operably attached to the front rail such that the first panel and rear rail move together as a unit relative to the front rail and the second panel to adjust spacing between the front and rear edges along the adjustment axis.

(13) In one example, the longitudinally adjustable rail includes a first rail member and a second rail member. The first rail member is attached to the front rail. The second rail member is attached to the rear rail. The second rail member is slidably connected to the first rail member to adjust the length of the longitudinally adjustable rail along the adjustment axis. The first panel is operably attached to the rear rail by being directly attached to the second rail member. The second panel is operably attached to the front rail by being directly attached to the first rail member.

(14) In one example, the first and second panels are removable from the frame simultaneously as a single unit.

(15) In one example, a first panel connector connects the first panel to a rear portion of the adjustable frame. The interaction between the first panel connector and the rear portion of the adjustable frame fixes the first panel relative to the rear portion of the adjustable frame along the adjustment axis. A second panel connector connects the second panel to a front portion of the adjustable frame. The interaction between the first connector and the front portion of the adjustable frame fixes the second panel relative to the front portion of the adjustable frame along the adjustment axis. The front and rear portions of the adjustable frame being movable relative to one another along the adjustment axis.

(16) In one example, the first panel connector is provided by at least one of a pin, a plug, and/or a projection formed by the first panel. The second panel connector is provided by at least one of a pin, a plug, and/or a projection formed by the second panel.

(17) In one example, the first panel includes at least one mounting projection extending downward below the top surface. The at least one mounting projection extends into a mounting aperture formed in a rear portion of the adjustable frame. The interaction between the at least one mounting projection and mounting aperture fixes the first panel relative to the rear portion of the adjustable frame along the adjustment axis. The second panel includes at least one mounting projection extending downward below the top surface. The at least one mounting projection extends into a mounting aperture formed in a front portion of the adjustable frame. The interaction between the at least one mounting projection and mounting aperture fixes the second panel relative to the front portion of the adjustable frame along the adjustment axis. The front and rear portions of the adjustable frame being movable relative to one another along the adjustment axis.

(18) In one example, the rear portion of the adjustable frame is movable relative to the front portion of the adjustable frame along the adjustment axis.

(19) In one example, the first panel includes at least one mounting projection that extends downward below the top surface. The at least one mounting projection extends into a mounting

aperture formed in the rear rail. The interaction between the at least one mounting projection and mounting aperture fixes the first panel relative to the rear rail along the adjustment axis. The second panel includes at least one mounting projection that extends downward below the top surface. The at least one mounting projection extends into a mounting aperture formed in the front rail. The interaction between the at least one mounting projection and mounting aperture fixing the second panel relative to the front rail along the adjustment axis.

(20) In one example, the at least one mounting projection of the first panel is formed by an annular wall that surrounds an aperture extending through the first panel and through the top surface. The at least one mounting projection of the second panel is formed by an annular wall that surrounds an aperture extending through the second panel and through the top surface.

(21) In one example, the at least one mounting projections of the first and second panels are sized for receipt of one or more fingers of a user to allow the user to grasp the corresponding first or second panel to remove the adjustable rack from the adjustable frame.

(22) In one example, an intermediate rail extends between and interconnects the first and second side rails. The second panel includes a channel formed in an underside thereof that receives the intermediate rail.

(23) In one example, the first and second panels have a sliding interlock therebetween. When coupled, the sliding interlock permits the first panel to slide relative to the second panel along the adjustment axis but prevents the first and second panels from being disconnected from one another in a direction generally orthogonal to the top surface.

(24) In one example, the sliding interlock includes at least one clip provided one of the first and second panels. The sliding interlock includes a slide region of the other one of the first and second panels that is slidably engaged by the at least one clip. The slide region being generally parallel to the adjustment axis.

(25) In one example, the slide region includes a notch sized to receive the at least one clip therethrough generally orthogonal to the top surface for assembling the sliding interlock.

(26) In one example, the sliding interlock includes first and second clips provided by one of the first and second panels. The first and second clips are spaced apart from one another in a lateral direction being generally perpendicular to the adjustment axis. First and second slide regions are provided by the other one of the first and second panels. The first clip slidably engages the first slide region for sliding motion parallel to the adjustment axis but prevents disconnection generally orthogonal to the top surface. The second clip slidably engages the second slide region for sliding motion parallel to the adjustment axis but prevents disconnection generally orthogonal to the top surface.

(27) In one example, the sliding interlock includes first and second clips provided by the first panel. Each of the first and second clips includes a leg extending downward from a first main panel section of the first panel and a foot portion extending laterally from the leg. The foot and a bottom of the first main panel section forming a channel therebetween. First and second slide regions are provided by a second main panel section of the second panel. The first slide region is received in the channel formed by the first clip. The second slide region is received in the channel formed by the second clip.

(28) In one example, the foot of the first clip extends towards the foot of the second clip. The first and second slide regions are positioned laterally between the legs of the first and second clips.

(29) In one example, the second main panel includes a first notch adjacent the first slide region and a second notch adjacent the second slide region. The first and second notches are sized and configured such that the foot of the first clip can pass through the first notch to align the first slide region with the channel of the first clip and the foot of the second clip can pass through the second notch to align the second slide region with the channel of the second clip.

(30) In one example, the first notch is formed between opposed ends of the first slide region such that a first portion of the first slide region is on a first side of the first notch and a second portion of

the first slide region is on a second side of the first notch. The second notch is formed between opposed ends of the second slide region such that a first portion of the second slide region is on a first side of the second notch and a second portion of the second slide region is on a second side of the first notch.

(31) In one example, the adjustable frame includes a first rear frame mount. The first rear frame mount includes an exposed portion that extends rearward of a rear side of the rear rail. The exposed portion includes a downward opening notch. The adjustable frame includes a first front frame mount. The first front frame mount includes an exposed portion that extends forward of a front side of the front rail. The exposed portion includes a downward opening notch.

(32) In one example at least two rear frame mounts or at least two front frame mounts are provided such that at least three frame mounts are provided to improve stability of the adjustable rack system, when mounted to the shelving frame.

(33) In one example, the first rear frame mount is removably attached to the rear rail. The first rear frame mount includes a first leg and a second leg extending from the first leg forming an L-shape. The first leg is longer than the second leg. The downward opening notch is formed in the exposed portion of the first leg. The first leg has a second portion located forward of the rear side of the rear rail. The second portion, at least in part, secures the first rear frame mount to the rear rail. The first front frame mount is removably attached to the front rail. The first front frame mount includes a first leg and a second leg extending from the first leg forming an L-shape. The first leg is longer than the second leg. The downward opening notch is formed in the exposed portion of the first leg. The first leg has a second portion located rearward of the front side of the front rail. The second portion, at least in part, secures the first front frame mount to the front rail.

(34) In one example, the rear rail includes a top wall, a bottom wall vertically spaced from the top wall, and a rear wall below the top wall. The rear wall defines the rear side of the rear rail. The rear wall defines an opening through which the first leg of the first rear frame mount extends. The bottom wall defines an opening through which the second leg of the first rear frame mount extends. The front rail includes a top wall, a bottom wall vertically spaced from the top wall, and a front wall below the top wall. The front wall defines the front side of the front rail, the front wall defines an opening through which the first leg of the first front frame mount extends. The bottom wall defines an opening through which the second leg of the first front frame mount extends.

(35) In one example, the rear rail includes a yoke defining a cavity that receives the second portion of the first leg of the first rear frame mount. The yoke is positioned between the rear wall and the opening in the bottom wall of the rear rail. The opening in the bottom wall of the rear rail is elongated in a lateral direction and sized to permit the first rear frame mount to be rotated from an orientation wherein the second leg is generally horizontal to an orientation wherein the second leg of the first rear frame mount extends vertically downward to insert the second leg of the of the first rear frame mount into the opening in the bottom wall. Insertion of the second leg of the first rear frame mount into the opening in the bottom wall limits motion of the first rear frame mount parallel to the adjustment axis. The front rail includes a yoke defining a cavity that receives the second portion of the first leg of the first front frame mount. The yoke is positioned between the front wall and the opening in the bottom wall of the front rail. The opening in the bottom wall of the front rail is elongated in a lateral direction and sized to permit the first front frame mount to be rotated from an orientation wherein the second leg of the first front frame mount is generally horizontal to an orientation wherein the second leg of the first front frame mount extends vertically downward to insert the second leg of the of the first front frame mount into the opening in the bottom wall. Insertion of the second leg of the first front frame mount into the opening in the bottom wall limits motion of the first rear frame mount parallel to the adjustment axis.

(36) In one example, the yoke of the rear rail extends downward from the top wall of the rear rail and the yoke of the front rail extends downward from the top wall of the front rail.

(37) In one example, the rear rail includes a rear wall. The rear wall defines an opening through

which the first rear frame mount extends. The first rear frame mount includes an upward opening notch. The upward opening notch receives a portion of the rear wall forming part of the opening through which the rear frame mount extends. The front rail includes a front wall. The front wall defines an opening through which the first front frame mount extends. The first front frame mount includes an upward opening notch. The upward opening notch receives a portion of the front wall forming part of the opening through which the front frame mount extends.

(38) In one example, the first rear frame mount includes an upward opening notch. The upward opening notch receives a portion of the rear rail. Engagement provided by the notch and the portion of the rear rail prevents movement of the first rear frame mount relative to the rear rail along the adjustment axis. The first front frame mount includes an upward opening notch. The upward opening notch receives a portion of the front rail. Engagement provided by the notch and the portion of the front rail prevents movement of the first front frame mount relative to the front rail along the adjustment axis.

(39) In one example, the top surface of the adjustable rack includes a plurality of upward extending ribs and a plurality of apertures formed therethrough.

(40) In one example, the plurality of upward extending ribs are provided by the first panel and the plurality of apertures are provided by the second panel.

(41) In an example, a frame mount for mounting a rack frame to a shelving unit frame is provided. The frame mount includes a first leg and a second leg extending from the first leg in a first direction forming an L-shape. The first leg is longer than the second leg. A first notch is formed in first leg. The notch opens in the first direction.

(42) In one example, the first and second legs are formed from a continuous piece of material.

(43) In one example, the continuous piece of material is a bent wire.

(44) In one example, a second notch is provided that opens in a second direction being opposite the first direction such that the first and second notches are on opposite sides of the first leg.

(45) In one example, the second notch is located axially along the first leg between the first notch and the second leg.

(46) In one example, the first notch extends axially along the first leg a greater distance than the second notch.

(47) In one example, the second notch is axially offset from the first notch along a length of the first leg.

(48) In an example, a shelving system including a shelving frame an adjustable rack system as outlined above is provided. The shelving frame includes a rear mounting wall and a front mounting wall. The rear mounting wall extends vertically a first height and the front mounting wall extends vertically a second height that is less than the first height. The front mounting wall is spaced forward of the rear mounting wall. The rear mounting wall and front mounting wall form a well therebetween below a top end of the front mounting wall. The adjustable rack is mounted to the rear mounting wall and the front mounting wall with the front edge of the adjustable rack at or below the top end of the front mounting wall of the shelving frame.

(49) In one example, the adjustable rack of the adjustable rack system is removable from the shelving frame without removing the adjustable frame from the shelving frame.

(50) In one example, a shelf arrangement providing a product support surface is mounted to the rear mounting wall. The product support surface is located vertically above the top end of the front wall.

(51) In one example, the front edge of the adjustable rack is positioned a first distance from the rear mounting wall. A front edge of the shelf arrangement is positioned a second distance from the rear mounting wall.

(52) In one example, the rear mounting wall includes a first plurality of vertically spaced mounting apertures. The front mounting wall includes a second plurality of vertically spaced mounting apertures. The adjustable rack system is selectively mountable within the first and second plurality

of vertically spaced mounting apertures to adjust a height and/or slant of the adjustable rack system relative to the shelving frame.

(53) In one example, a shelf arrangement providing a product support surface. The product support surface is located vertically above the top end of the front wall. The rear mounting wall includes a third plurality of vertically spaced mounting apertures that is different than the first plurality of mounting apertures. The shelf arrangement is mounted to the third plurality of vertically spaced mounting apertures.

(54) In one example, the third plurality of vertically spaced mounting apertures includes apertures that are vertically above all of the apertures of the first and second plurality of vertically spaced mounting apertures.

(55) Other aspects, objectives and advantages of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings incorporated in and forming a part of the specification illustrate several aspects of the present invention and, together with the description, serve to explain the principles of the invention. In the drawings:

(2) FIG. 1 is a cross-sectional perspective view of a retail merchandise display in the form of a shelving system according to an example of the invention;

(3) FIG. 2 is a cross-sectional illustration of the shelving system of FIG. 1;

(4) FIG. 3 is an enlarged partial cross-sectional illustration of the well of the shelving system of FIG. 1;

(5) FIG. 4 is cross-sectional and perspective illustration of the shelving frame of the shelving system of FIG. 1;

(6) FIG. 5 is a partial cross-sectional illustration of the shelving system of FIG. 1;

(7) FIG. 6 is a perspective illustration of an adjustable rack system of the shelving system of FIG. 1;

(8) FIG. 7 is an exploded illustration of the adjustable rack system of FIG. 6;

(9) FIG. 8 is a partial exploded illustration of adjustable rack system of FIG. 6 having the adjustable rack removed from the adjustable frame;

(10) FIG. 9 is a cross-sectional illustration of the adjustable rack of the adjustable rack system of FIG. 6;

(11) FIG. 10 is a bottom view illustration of the adjustable rack of the adjustable rack system of FIG. 6;

(12) FIG. 11 is an enlarged perspective illustration of the adjustable rack of the adjustable rack system of FIG. 6 illustrating a portion of the sliding interlock between the panels of the adjustable rack;

(13) FIG. 12 is a bottom perspective illustration of a first panel of the adjustable rack of the adjustable rack system of FIG. 6;

(14) FIG. 13 is a top perspective illustration of a first panel of the adjustable rack of the adjustable rack system of FIG. 6;

(15) FIG. 14 is a top perspective illustration of a second panel of the adjustable rack of the adjustable rack system of FIG. 6;

(16) FIG. 15 is a bottom perspective illustration of a second panel of the adjustable rack of the adjustable rack system of FIG. 6;

(17) FIG. 16 is a cross-sectional illustration of the adjustable rack system of FIG. 6 taken about line 16-16 in FIG. 6;

- (18) FIG. 17 is a cross-sectional illustration of the adjustable rack system of FIG. 6 taken about line 17-17 in FIG. 6 taken through the rear rail;
- (19) FIG. 18 is a cross-sectional illustration of the adjustable rack system of FIG. 6 taken about line 18-18 in FIG. 6 taken through the front rail;
- (20) FIG. 19 is a cross-sectional illustration of a laterally extending rail (e.g. rear rail or front rail) and an installed frame mount (e.g. front or rear frame mount);
- (21) FIG. 20 is a partial cross-sectional illustration of the shelving system of FIG. 1 illustrating the mounting of a shelving arrangement and the adjustable rack to the rear mounting wall of the shelving frame;
- (22) FIG. 21 is a perspective view of a laterally extending rail (e.g. rear rail or front rail);
- (23) FIG. 22 is a simplified illustration of a later extending rail (e.g. rear rail or front rail) and an installed frame mount (e.g. front or rear frame mount), with the frame mount in the installed position; and
- (24) FIG. 23 is a simplified illustration of a later extending rail (e.g. rear rail or front rail) and an installed frame mount (e.g. front or rear frame mount), with the frame mount in a rotated position prior to being fully installed.
- (25) While the invention will be described in connection with certain preferred embodiments, there is no intent to limit it to those embodiments. On the contrary, the intent is to cover all alternatives, modifications and equivalents as included within the spirit and scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE INVENTION

- (26) FIG. 1 illustrates an example of a retail merchandise display in the form of a shelving system **100**. The shelving system **100** may take the form of a refrigerated case for displaying retail merchandise such as produce or other food items that are to be cooled including, but not limited to yogurt, cheeses, milk, beverages, etc. The shelving system **100** includes a shelving frame **102** that supports one or more shelf arrangements **104** and one or more adjustable rack systems **106**.
- (27) The shelving frame **102** includes a rear mounting wall **108** and a front mounting wall **110** that is spaced apart from the rear mounting wall **108**. The shelf arrangement **104** is mounted in a cantilevered fashion from the shelving frame **102** and the adjustable rack system **106** is mounted to and between the rear mounting wall **108** and front mounting wall **110**. The shelf arrangement **104** provides a product support surface upon which products are displayed.
- (28) With additional reference to FIG. 2, the shelving system **100**, in this example, defines a well **112** between the rear and front mounting walls **108**, **110**. The well **112** includes a bottom **113**. In some embodiments, a set of plenums may be interposed between the bottom **113** and the adjustable rack system **106**.
- (29) The rear mounting wall **108** includes a plurality of vertically spaced rack mounting apertures **114**. With reference to FIG. 4, in this example, the rack mounting apertures **114** in the rear mounting wall **108** are grouped in vertical sets of three rack mounting apertures **114**. Similarly, the front mounting wall **110** includes a plurality of vertically spaced rack mounting apertures **116** (see e.g. FIG. 3). In this example, the rack mounting apertures **116** in the front mounting wall are grouped in vertical sets of three rack mounting apertures **116**.
- (30) With reference to FIGS. 3 and 5, the adjustable rack systems **106** are mountable within selective ones of the rack mounting apertures **114**, **116** to adjust the vertical position and angled orientation of the adjustable rack system **106** within the shelving frame **102**. For example, the maximum angle can be provided by mounting the adjustable rack system **106** to the upper most mounting apertures **114** in the rear mounting wall **108** and to the lower most mounting apertures **116** in the front mounting wall **110**. Further, the maximum height can be provided by mounting the adjustable rack system **106** to the upper most mounting apertures **114**, **116** or the lower most height can be provided by mounting the adjustable rack system **106** to the lower most mounting apertures **114**, **116**. Vertical adjustability of the front and rear edges of the adjustable rack system **106** is

represented by arrows **115**, **117** in FIGS. **3** and **5**.

(31) In addition to mounting apertures **114** for mounting the adjustable rack systems **106**, the rear mounting wall **108** includes a plurality of vertically spaced apart shelf mounting apertures **120**. Mounting features, such as hooks, of the shelf arrangements **104** are used to selectively mount the shelf arrangements **104** at desired heights using the shelf mounting apertures **120**. While only a single shelf arrangement **104** is illustrated in FIG. **1**, multiple vertically spaced apart shelf arrangements **104** can be provided.

(32) In this example, the shelving frame **102** includes outer most rear and front walls **122**, **124**. The rear and front mounting walls **108**, **110** are inward of these outer most walls **122**, **124**. Air cavity **126** is formed between the rear mounting wall **108** and the rear outer most wall **122** and air cavity **128** is formed between the front most mounting wall **110** and the front outer most wall **124**.

(33) Further, in some embodiments, the rear mounting wall **108** may be formed from a plurality of components. For example, rear mounting wall **108** could include a plurality of vertical upright members that provides the shelf mounting apertures **120**. One or more wall panels could be mounted between the vertical upright members and provide the mounting apertures **114** for mounting the adjustable rack systems **106**. Alternatively, all of the mounting apertures **114**, **120** could be formed in a single panel that provides the rear mounting wall **108**.

(34) As illustrated in FIG. **2**, the rear mounting wall **108** extends vertically a first height H1, while the front mounting **110** wall extends vertically a second height H2. The height H2 of the front mounting wall **110** is less than the height H1 of the rear mounting wall. Again, the front and rear mounting walls **110**, **108** are spaced apart from one another and the bottom **113** is below a top end **130** of the front mounting wall **110** forming well **112**.

(35) When assembled, the shelf arrangements **104** are typically located vertically above the top end **130** of the front mounting wall **110** and vertically above the adjustable rack system **106**.

(36) The adjustable rack system **106** extends between a rear edge **132** and a front edge **134**. In this example, the adjustable rack system **106** is mounted within the well **112** such that, at least, the front edge **134** is positioned vertically at or below the top end **130** of the front mounting wall **110** of the shelving frame **102**.

(37) Typically, the adjustable rack system **106** will extend forward of the front edge of the shelf arrangements **104**. More particularly, the shelf arrangements **104** will extend forward of the rear mounting wall **108** a shorter distance than the amount that the adjustable rack system **106** will extend forward of the rear mounting wall **108**. As illustrated in FIG. **1**, accessories such as false rear walls **135** may be provided that extend within the vertical gap formed between the bottom of the shelf arrangement **104** and the top of the adjustable rack system **106**. The false rear walls **135** may be movable forward and backward relative to the adjustable rack system depending on the amount of product to be displayed in front of the false rear walls **135**. Further yet, dividers **137** may be provided that are mounted to the adjustable rack system **106** to provide distinct storage regions above the adjustable rack system **106**.

(38) In the illustrated shelving system **100** of FIG. **1**, product can be stored within the well **112** on the adjustable rack system **106**. Typically, product will be stored on the front panel **174**, but could be stored on the rear panel **172** as well. The apertures in the front panel **174** assist in vertical air flow through the adjustable rack system **106**.

(39) With additional reference to FIGS. **6** and **7**, the adjustable rack system **106** includes an adjustable frame **140** and an adjustable rack **142** removably attached to the adjustable frame **140**.

(40) With principle reference to FIG. **7**, the adjustable frame **140** includes a front rail **144**, a rear rail **146**, a pair of adjustable longitudinal rails in the form of first side rail **148** and a second side rail **150**. The adjustable frame **140**, in this example, also includes a pair of intermediate rails **152**, **154**. The first and second side rails **148**, **150** interconnect the front and rear rails **144**, **146** while the front and rear rails **144**, **146** maintain the lateral spacing between the first and second side rails **148**, **150**.

(41) The length L1 of the adjustable rack system **106** and spacing between the rear and front edges **132, 134** is adjustable along adjustment axis **155**, as illustrated by arrow **156**. This allows the adjustable rack system **106** to be used in different shelving frames (e.g. where the rear and front mounting walls **108, 110** are closer or farther apart than in the illustrated example). Further, this allows for adjustment in the length L1 of the adjustable rack system **106** depending on the necessary length when adjusting the angle of the adjustable rack system **106** within a given shelving frame **102**.

(42) To allow for adjusting the length L1 of the adjustable rack system **106**, both the adjustable frame **140** and the adjustable rack **142** are likewise lengthwise adjustable.

(43) To permit for adjusting the length of the adjustable frame **140**, the first and second side rails **148, 150** are lengthwise adjustable. Adjusting the length of the first and second side rails **148, 150** adjusts the spacing between the front and rear rails **144, 146** along the adjustment axis **155**. While a pair of side rails **148, 150** are illustrated, other examples may include more or less longitudinally adjustable rails. For instance, a single central rail extending generally parallel to the adjustment axis could be included.

(44) Each of the first and second side rails **148, 150** includes a front side rail member **160** and a rear side rail member **162**. For each side rail **148, 150**, the front side rail member **160** is slidably engaged with the rear side rail member **162** such that the front side rail member **160** is adjustably positionable relative to the rear side rail member **162** to adjust the length of the corresponding side rail **148, 150**. The front rail **144** interconnects the two front side rail members **160**. The rear rail **146** interconnects the two rear side rail members **162**.

(45) As such, the front rail **144** and two front side rail members **160** form a first unit that defines a front portion of the adjustable frame that moves relative to a second unit formed by the rear rail **146** and two rear side rail members **162** that defines a rear portion of the adjustable frame.

(46) In this example, the two front side rail members **160** slide into and between the two rear side rail members. In one example, (with reference to FIG. 8) each rear side rail member **162** forms one or more channels **164** between inner and outer walls **166, 168** that receive the corresponding front side rail member **160**. The inner and outer walls **166, 168** are connected by a top wall **170**.

(47) The adjustable rack **142** is formed by a plurality of interconnected panels that can slide relative to one another along the adjustment axis **155** so as to allow for adjusting the length of the adjustable rack **142** as well as spacing between rear and front edges **132, 134**. In this example, the adjustable rack **142** includes a first panel **172** and a second panel **174**. The first panel **172** is a rear most panel and the second panel **174** is the front most panel. The first panel **172** defines rear edge **132** while the second panel **174** defines front edge **134**. The first and second panels **172, 174** are interconnected such that they are adjustably positional

(48) The first panel **172** extends over a rear most portion of the second panel **174**.

(49) The first and second panels **172, 174** combine to form a top surface **176** that is formed in part by each of the first and second panels **172, 174**.

(50) In this example, the top surface **176** includes a plurality of upward extending ribs **178** as well as a plurality of apertures **180**. In this example, first panel **172** includes the ribs **178** while the second panel **174** includes apertures **180**. In this example, the ribs **178** are generally parallel to side rails **148, 150** and adjustment axis **155**. Further, the apertures **180** extend entirely through second panel **174**.

(51) The two side edges and rear edge of the first panel **172**, in this example are raised, but need not be in other examples. Further, the front edge and portions of the two side edges of the second panel **174** are raised in this example.

(52) The length L2 of the first panel **172** is shorter than the length L3 of the second panel **174**.

(53) With reference to FIGS. 8-11, the first and second panels **172, 174** are coupled by a sliding interlock **180** therebetween. The sliding interlock permits the first and second panels **172, 174** to slide relative to one another along the adjustment axis **155** but prevents the first and second panels

from being disconnected from one another along an axis generally orthogonal to the top surface of the adjustable rack **142**.

(54) The sliding interlock **180** includes first and second clips **182** provided by the first panel **172** and a pair of slide regions **184** provided by the second panel **174**. The slide regions **184** are slidably received in channels **186** formed by the clips **182**.

(55) The clips **182**, in this example, are unitarily formed with the rest of the first panel **172** as a single continuous piece of material.

(56) Each clip **182** includes a leg portion **188** that extends downward from a main panel **190** of first panel **172**. A foot portion **192** extends laterally, inward in this example, from the leg portion **188**. The gap between the bottom of main panel **190** and a top surface of foot **192** forms the channel **186**. The foot **192** thus forms an undercut region. The foot **192** and main panel **190** bound slide region **184** to prevent disengagement in a direction orthogonal to the top surface of the adjustable rack **142** while allowing sliding motion parallel to adjustment axis **155**.

(57) In this example, the slide regions **184** are formed as part of edges defining the outer most periphery of the second panel **174**. However, in other embodiments, the slide regions could be inboard of the outer most periphery and could be provided by slots formed in the second panel **174**. In such an arrangement

(58) In this example, the feet **192** of the pair of clips **182** extend toward one another in this example. The slide regions **184** are thus located laterally between the pair of legs **188** of the pair of clips **182**.

(59) An abutment stop **194** is provided at one end of the slide regions **184** to limit axial sliding motion of the sliding interlock and to prevent unexpected disconnect of the first and second panels **172**, **174** when adjusting the axial length of the adjustable rack **142**. If the user attempts to over extend the adjustable rack **142**, an end clip **182** will abut stop **194** limiting relative motion between the panels **172**, **174**.

(60) At an opposite end of each slide region **184** a notch **196** is provided in main panel **193** of the second panel **174**. The notches **196** allow for assembly of the adjustable rack **142** and particularly for passage of the feet **192** past the slide regions **184**. Thus, the notches **196** are sized and configured to permit the feet **192** to pass therethrough. However, in other examples, the clips **182** may be sufficiently flexible that they clips can flex to allow for attachment of the clip **182** to the slide regions **184**.

(61) In this example, the surface area of the main panels **190**, **193** has less than 50% being apertures and in preferred embodiments less than 30% and even more preferably less than 25%. This reduces the difficulties in cleaning the panels **172**, **174** as compared to wire racks.

(62) With reference to FIG. **10**, a width **W1** between the inner most edge of the notches **196** is less than the width **W2** between the inner most edges of the feet **192**. Similarly, the width **W3** between the outer most portion of slide regions **184** is greater than the width **W2** between the inner most edges of the feet **192**.

(63) In this example, at the end of each notch **196**, opposite slide regions **184**, is an abutment **198** that assists in quickly aligning the clips **182** with the notches **196** when disassembling the adjustable rack **142**. The width **W4** provided by the abutments **198** is greater than width **W1** and width **W3** so that the clips **182** will abut the abutments **198**. The width **W5** of abutments **194** is greater than width **W3**.

(64) While discussed as having the clips **182** be part of the first panel **172** and the slide regions **184** be part of the second panel **174**, it is contemplated that the features could be switched such that the clips **182** are part of second panel **174** and the slide regions **184** are part of the first panel **172**.

(65) In other embodiments, the notches **196** could be formed in between portions of a slide region **184** such that portions of the slide region straddle the notches **196**.

(66) FIGS. **12-15** are further illustrations of the first and second panel **172**, **174**.

(67) With reference to FIGS. **16-18**, the adjustable rack **142** is removably attached to the adjustable

frame **140**. The first panel **172** is operably attached to the rear rail **146** such that the first panel **172** and rear rail **146** move as a unit together relative to the front rail **144** and second panel **174** along the adjustment axis **155**. This allows for adjusting a length of the adjustable rack system **106**.

(68) The adjustable rack **142** may be removed from the adjustable frame **140** simultaneously as a single unit.

(69) To removably attach the adjustable rack **142** to the adjustable frame **140**, the first panel **172** includes a plurality of mounting projections **200** that extend downward from main panel **190** and below the top surface defined by the adjustable rack **142**. The mounting projections **200** extend into corresponding mounting apertures **202** formed in the rear rail **146**. The interaction between the mounting projections **200** and mounting apertures **202** fixes the first panel **172** relative to the rear rail **146** along the adjustment axis **155** such that the two components move together and prevent movement therebetween along the adjustment axis **155**. However, by being axially inserted into the apertures **202**, the first panel **172** can be disconnected from the rear rail **146** along an axis **205** perpendicular to the top surface and main panel **190**.

(70) Similarly, the second panel **174** includes a plurality of mounting projections **206** that extend downward from main panel **193** and below the top surface defined by the adjustable rack **142**. The mounting projections **206** extend axially into corresponding mounting apertures **208** formed in the front rail **144**. The interaction between the mounting projections **206** and the mounting apertures **208** fixes the second panel **174** to the front rail **144** along the adjustment axis **155** such that the two components move together and prevent movement therebetween along the adjustment axis **155**. However, by having the projections **206** axially inserted into the apertures **208**, the second panel **174** can be disconnected from the front rail **144** along axis **205** perpendicular to the top surface and main panel **190**.

(71) In this example, the mounting projections **200**, **206** are formed by annular walls that extend axially downward from main panels **190**, **193**, respectively. The annular walls forming the mounting projections **200**, **206** define apertures **210**, **212** that extend through the mounting projections **200**, **206**. The apertures **210**, **212** extend through the top surface of the panels **172**, **174** and the top surface of the adjustable rack **142**. The apertures **210**, **212** are sized and configured for receipt of a user's finger or fingers to allow a user to grasp the first and second panels **172**, **174** to remove the adjustable rack **142** axially upward from adjustable frame **140**.

(72) While the mounting apertures **202**, **208** are illustrated as being formed in the front and rear rails **144**, **146**, in other embodiments, the front and rear mounting apertures **202**, **208** could be formed in the front and/or rear side rail members **160**, **162** rather than in the front and rear rails **144**, **146**. In such an arrangement, the panels **172**, **174** would be operably connected to the front and rear rails **144**, **146**, albeit indirectly by way of the front and rear side rail members **160**, **162**.

(73) With reference to FIGS. **8**, **10** and **16**, the adjustable frame **140** includes intermediate rail **152** that extends laterally between the first and second side rails **148**, **150** and particularly the front side rail members **160** thereof. The second panel **174** includes a channel **216** that axially receives the intermediate rail **152**. The channel **216** further axially fixes the second panel **174** relative to front rail **144**. The channel **216** is formed in an underside of second panel **174**.

(74) The intermediate rail **152** also vertically supports the inner portion (e.g. portion opposite front edge **134**) of the second panel **174**.

(75) With reference to FIGS. **3** and **5-7**, the adjustable rack system, and particularly adjustable frame **140** thereof, is removably mountable to the shelving frame **102** and particularly the rear and front mounting walls **108**, **110** as noted previously.

(76) Rear and front frame mounts **220**, **222** mount the adjustable frame **140** to the shelving frame **102**. The rear and front frame mounts **220**, **222** are identical in the illustrated embodiment, except the rear frame mounts **220** are operably attached to the rear rail **146** while the front frame mounts **222** are operably attached to the front rail **144**.

(77) As illustrated in FIG. **19**, a rear frame mount **220** is mounted removably within rear rail **146**.

Notably, the mounting of the front frame mounts **222** to the front rail **144** is the same.

(78) The rear frame mount **220** in this example includes a first leg **226** and a second leg **228**. The first leg **226** has a length **L2** that is greater than a length **L3** of the second leg **228**. The second leg **228** extends from the first leg **226** forming an L-shape. In this example, the rear frame mount **220** is a single continuous piece of material. The rear frame mount **220** can be formed from a bent wire.

(79) When installed, the rear frame mount **220** includes an exposed portion **230** and a mounting portion **232**. The exposed portion **230** is external to the rear rail **146**. In particular, the exposed portion **230** extends rearward of a rear side **234** of the rear rail **146**. The mounting portion **232** is forward of the rear side **234** and located generally below top wall **236** of the rear rail **146**. The mounting portion **232** secures the rear frame mount **220**, at least in part, to the rear rail **146**.

(80) The exposed portion **230** of the first leg **226** mates with the rear mounting wall **108** of the shelving frame **102** (see e.g. FIGS. **5** and **20**). The exposed portion **230** includes a downward opening notch **238**. The exposed portion **230** extends through a selected rack mounting aperture **114** with the downward opening notch **238** engaging the rear mounting wall **108**. This helps properly locate and secure the adjustable frame **140** to the rear mounting wall **108**. The notch region of the first leg **226** will rest on the portion of the rear mounting wall **108** that forms the bottom portion of mounting aperture **114**.

(81) Notably, the length **L4** of the notch **238** is sufficiently greater than the thickness **T1** of the rear mounting wall **108**. This allows for the notch **238** to be more easily installed as well as to accommodate changes in angle of the adjustable rack system **106** as discussed previously. In some examples, Length **L4** is at least twice thickness **T1**, at least three times thickness **T1**, and more preferably at least four times thickness **T1**.

(82) As illustrated in FIG. **19**, the first leg **226** includes an upward opening notch **240**. The upward opening notch **240** receives a portion of a rear wall **242** of the rear rail **146**. This portion of the rear wall **242** bounds an opening **244** through which the first leg **226** extends.

(83) The rear rail **146**, as noted above, has a top wall **236** and a rear wall **242**. The rear rail also includes a front wall **246** and a bottom wall **248**. The bottom wall **248** is connected to the top wall **236** by front wall **246**. The bottom wall **248** includes a laterally elongated aperture **250** that receives the second leg **228** axially therethrough. One or more of the engagement between notch **240** and rear wall **242** and the engagement of the second leg **228** and the bottom wall **248** axially locates the rear frame mount **220** relative to the rear rail **146**.

(84) Notch **238** extends along the first leg **226** a further length than notch **240**.

(85) FIGS. **21-23** illustrate how aperture **250** is laterally elongated. Notably, the length **L5** of aperture **250** is such that the rear frame mount **220** can be rotated about axis **255** from an orientation where second leg **228** is generally horizontal and not extending into aperture **250** (see FIG. **23**) to an orientation where second leg **228** is substantially vertical and extends into aperture **250** (see FIG. **22**).

(86) Aperture **244**, in the illustrated embodiment, is generally D-shaped and has a flat region **260** that is received in upward opening notch **240**.

(87) Rear rail **146** further includes a downward opening yoke **262** that includes a cavity that receives the mounting portion **232** of the first leg **226**. The yoke **262** extends downward from the top wall **236** and is located between the front and rear walls **246**, **242**. The yoke **262** is also located between the rear wall **246** and aperture **250** in the bottom wall **250**.

(88) Rear rail **146** and particularly front wall **246** may include a further elongated aperture **266** to assist in installation of the rear frame mount **220**. This elongated aperture **266** is sized to allow second leg **228** to pass therethrough when installing the rear frame mount **220**.

(89) Elongated apertures **250**, **266** are generally laterally elongated perpendicular to the adjustment axis **155**. They are also preferably laterally aligned with one another. Further yet, apertures **244**, **250**, **266** and the opening of yoke **262** are, at least in part, axially aligned as illustrated by axis **268**.

(90) In this example notch **240** is positioned between notch **238** and second leg **228**. Further, notch

240 is axially offset from notch **238** along the first leg **226**.

(91) In one example, the rear rail **146** is formed from stamped and bent sheet metal.

(92) While a set of rear frame mounts **220** and rear rail **146** were described, the same exact explanation applies to the front frame mounts **222** and the front rail **144**. However, instead extending rearward beyond a rear side of the rear wall of the rear rail **146**, the front frame mounts **222** extend forward of a front side of a front wall of the front rail **144**. Further, the front rail **144** is substantially identical to the rear rail **146** except it may be a slight bit shorter as it located between the two front side rail members **160** rather than the two rear side rail members **168**.

(93) In operation, the adjustable rack system **106** may be assembled and then lengthwise adjusted when installing into the shelving rack **102**. Alternatively, the adjustable frame **140** can be installed and then the adjustable rack **142** can be installed on the adjustable frame **140**.

(94) Once the adjustable frame **140** is installed, it need not be removed from the shelving frame **102** and well **112** thereof to provide access to the portion of the well **112** located below the adjustable rack system **106**, such as to the bottom **113**. Access may be required for cleaning purposes or maintenance purposes, such as to accesses other componentry such as if the shelving system **100** is used as a refrigerated display or a display that includes misters or air curtains (e.g. pumps, valves, controllers, fluid supply lines, cooling units, cooling fluid supply lines, etc.).

(95) Here, the first and second panels **172**, **174** can be removed simultaneously from the adjustable frame **140** and without requiring removal of the adjustable frame. This makes the components that are removed, e.g. panels **172**, **174**, much lighter than prior configurations. Further, by leaving the adjustable frame **140** in place within the shelving frame **102**, the proper orientation and height for the adjustable rack system **106** is not inadvertently changed if a user fails to reinstall the frame **140** in the desired mounting apertures **114**, **116**.

(96) In some examples, the adjustable rack system **106** may be configured such that the first panel **172** can be tilted relative to the second panel **174** and the rear rail **146** such that the first panel **172** can be disconnected from the rear rail **146** while the sliding interlock **180** remains connected. This allows the first panel **172** to be slid toward front edge **134** along the second panel **174** without removing the second panel **174** from the front rail **144**. This can allow access to the area below the first panel **172** within the well **112** without requiring removal of the adjustable rack system **106**. This can be allowed by providing sufficient clearance within the channels formed by clips **182** and the slide regions **184** and/or providing the clips and/or slide regions **184** and/or main panels **190**, **193** with sufficient flexibility.

(97) In some systems **100** as outlined above, such as those that include plenums below the adjustable rack system **106**, methods of use include removing one adjustable rack **142** while removing the adjustable frame **140** installed. Thereafter, a plenum under the adjustable rack system **106** may be slid laterally to provide access to the portion of the well below the plenum. This can be done without entirely removing the entire rack system **106**.

(98) All references, including publications, patent applications, and patents cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

(99) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) is to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”)

provided herein, is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the invention. (100) Preferred embodiments of this invention are described herein, including the best mode known to the inventors for carrying out the invention. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the invention to be practiced otherwise than as specifically described herein. Accordingly, this invention includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the invention unless otherwise indicated herein or otherwise clearly contradicted by context.

Claims

1. An adjustable rack system for mounting to a frame assembly of a shelving unit, the adjustable rack system comprising: an adjustable frame having: a front rail; a rear rail being spaced apart from the front rail; at least one longitudinally adjustable rail interconnecting the front and rear rails, the at least one longitudinally adjustable rail being longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis; and an adjustable rack removably attached to the adjustable frame, the adjustable rack defines a top surface, the adjustable rack having: a first panel defining a rear edge of the adjustable rack; and a second panel defining a front edge of the adjustable rack, the second panel being adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack, the top surface is defined in part by each of the first and the second panel; wherein the at least one longitudinally adjustable rail includes a first and a second side rail interconnecting the front and rear rails, the second side rail being laterally spaced apart from the first side rail; and further comprising an intermediate rail extending between and interconnecting the first and second side rails, the second panel including a channel formed in an underside thereof, the intermediate rail received in the channel.
2. The adjustable rack system of claim 1, wherein: the first side rail includes a first front side rail member and a first rear side rail member, the first front side rail member is adjustably positionable relative to the first rear side rail member to adjust the length of the first side rail, the first front side rail member is attached to the front rail, and the first rear side rail member is attached to the rear rail; and the second side rail includes a second front side rail member and a second rear side rail member, the second front side rail member is adjustably positionable relative to the second rear side rail member to adjust the length of the second side rail, the second front side rail member is attached to the front rail, and the second rear side rail member is attached to the rear rail.
3. The adjustable rack system of claim 1, wherein the first panel is operably attached to the rear rail and the second panel is operably attached to the front rail such that the first panel and rear rail move together as a unit relative to the front rail and the second panel to adjust spacing between the front and rear edges along the adjustment axis.
4. The adjustable rack system of claim 3, wherein: each of the first and second side rails includes: a first rail member and a second rail member; the first rail member is attached to the front rail; the second rail member is attached to the rear rail; the second rail member is slidably connected to the first rail member to adjust the length of the longitudinally adjustable rail and the spacing between the front and rear rails along the adjustment axis; the first panel is operably attached to the rear rail by being directly attached to the second rail member; and the second panel is operably attached to the front rail by being directly attached to the first rail member.
5. The adjustable rack system of claim 1, wherein the first and second panels are removable from the frame simultaneously as a single unit.

6. The adjustable rack system of claim 1, wherein: a first panel connector connects the first panel to a rear portion of the adjustable frame, the interaction between the first panel connector and the rear portion of the adjustable frame fixing the first panel relative to the rear portion of the adjustable frame along the adjustment axis; and a second panel connector connects the second panel to a front portion of the adjustable frame, the interaction between the first connector and the front portion of the adjustable frame fixing the second panel relative to the front portion of the adjustable frame along the adjustment axis.

7. The adjustable rack system of claim 6, wherein: the first panel connector is provided by at least one of a pin, a plug, and/or a projection formed by the first panel; and the second panel connector is provided by at least one of a pin, a plug, and/or a projection formed by the second panel.

8. The adjustable rack system of claim 1, wherein the first panel includes at least one mounting projection extending downward below the top surface, the at least one mounting projection extending into a mounting aperture formed in a rear portion of the adjustable frame, the interaction between the at least one mounting projection and mounting aperture fixing the first panel relative to the rear portion of the adjustable frame along the adjustment axis; and the second panel includes at least one mounting projection extending downward below the top surface, the at least one mounting projection extending into a mounting aperture formed in a front portion of the adjustable frame, the interaction between the at least one mounting projection and mounting aperture fixing the second panel relative to the front portion of the adjustable frame along the adjustment axis.

9. The adjustable rack system of claim 1, wherein: the first panel includes at least one mounting projection extending downward below the top surface, the at least one mounting projection extending into a mounting aperture formed in the rear rail, the interaction between the at least one mounting projection and mounting aperture fixing the first panel relative to the rear rail along the adjustment axis; and the second panel includes at least one mounting projection extending downward below the top surface, the at least one mounting projection extending into a mounting aperture formed in the front rail, the interaction between the at least one mounting projection and mounting aperture fixing the second panel relative to the front rail along the adjustment axis.

10. The adjustable rack system of claim 1, wherein the first and second panels have a sliding interlock therebetween, when coupled, the sliding interlock permits the first panel to slide relative to the second panel along the adjustment axis but prevents the first and second panels from being disconnected from one another in a direction generally orthogonal to the top surface.

11. The adjustable rack system of claim 10, wherein the sliding interlock includes: at least one clip provided on one of the first and second panels; a slide region of the other one of the first and second panels that is slidably engaged by the at least one clip, the slide region being generally parallel to the adjustment axis.

12. The adjustable rack system of claim 10, wherein the sliding interlock includes: first and second clips provided by one of the first and second panels, the first and second clips being spaced apart from one another in a lateral direction being generally perpendicular to the adjustment axis; first and second slide regions provided by the other one of the first and second panels; the first clip slidably engaging the first slide region for sliding motion parallel to the adjustment axis but preventing disconnection generally orthogonal to the top surface; and the second clip slidably engaging the second slide region for sliding motion parallel to the adjustment axis but preventing disconnection generally orthogonal to the top surface.

13. The adjustable rack system of claim 10, wherein the sliding interlock includes: first and second clips provided by the first panel, each of the first and second clips including a leg extending downward from a first main panel section of the first panel and a foot portion extending laterally from the leg, the foot portion and a bottom of the first main panel section forming a channel therebetween; first and second slide regions provided by a second main panel section of the second panel, the first slide region received in the channel formed by the first clip and the second slide region received in the channel formed by the second clip.

14. The adjustable rack system of claim 1, wherein the top surface of the adjustable rack includes a plurality of upward extending ribs and a plurality of apertures formed therethrough.
15. A shelving system comprising: a shelving frame including: a rear mounting wall extending vertically a first height; a front mounting wall extending vertically a second height that is less than the first height, the front mounting wall being spaced forward of the rear mounting wall, the rear mounting wall and front mounting wall forming a well therebetween below a top end of the front mounting wall; and an adjustable rack system of claim 1 mounted to the rear mounting wall and the front mounting wall with the front edge of the adjustable rack at or below the top end of the front mounting wall of the shelving frame.
16. The shelving system of claim 15, wherein the adjustable rack of the adjustable rack system is removable from the shelving frame without removing the adjustable frame from the shelving frame; and further comprising a shelf arrangement providing a product support surface, the product support surface being located vertically above the top end of the front wall.
17. The shelving system of claim 15, wherein the rear mounting wall includes a first plurality of vertically spaced mounting apertures and the front mounting wall includes a second plurality of vertically spaced mounting apertures, the adjustable rack system being selectively mountable within the first and second plurality of vertically spaced mounting apertures to adjust a height and/or slant of the adjustable rack system relative to the shelving frame; further comprising a shelf arrangement providing a product support surface, the product support surface being located vertically above the top end of the front wall; wherein the rear mounting wall includes a third plurality of vertically spaced mounting apertures that is different than the first plurality of mounting apertures, the shelf arrangement being mounted to the third plurality of vertically spaced mounting apertures; wherein the third plurality of vertically spaced mounting apertures includes apertures that are vertically above all of the apertures of the first and second plurality of vertically spaced mounting apertures.
18. An adjustable rack system for mounting to a frame assembly of a shelving unit, the adjustable rack system comprising: an adjustable frame having: a front rail; a rear rail being spaced apart from the front rail; at least one longitudinally adjustable rail interconnecting the front and rear rails, the at least one longitudinally adjustable rail being longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis; and an adjustable rack removably attached to the adjustable frame, the adjustable rack defines a top surface, the adjustable rack having: a first panel defining a rear edge of the adjustable rack; and a second panel defining a front edge of the adjustable rack, the second panel being adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack, the top surface is defined in part by each of the first and the second panel; wherein: the first panel includes at least one mounting projection extending downward below the top surface, the at least one mounting projection extending into a mounting aperture formed in the rear rail, the interaction between the at least one mounting projection and mounting aperture fixing the first panel relative to the rear rail along the adjustment axis; and the second panel includes at least one mounting projection extending downward below the top surface, the at least one mounting projection extending into a mounting aperture formed in the front rail, the interaction between the at least one mounting projection and mounting aperture fixing the second panel relative to the front rail along the adjustment axis; wherein: the at least one mounting projection of the first panel is formed by an annular wall that surrounds an aperture extending through the first panel and through the top surface; and the at least one mounting projection of the second panel is formed by an annular wall that surrounds an aperture extending through the second panel and through the top surface.
19. The adjustable rack system of claim 18, wherein the at least one mounting projections of the first and second panels are sized for receipt of one or more fingers of a user to allow the user to grasp the corresponding first or second panel to remove the adjustable rack from the adjustable frame.

20. The adjustable rack system of claim 18, wherein: the adjustable frame includes a first rear frame mount, the first rear frame mount includes an exposed portion that extends rearward of a rear side of the rear rail, the exposed portion includes a downward opening notch; and the adjustable frame includes a first front frame mount, the first front frame mount includes an exposed portion that extends forward of a front side of the front rail, the exposed portion includes a downward opening notch.

21. The adjustable rack system of claim 20, wherein: the rear rail includes a rear wall, the rear wall defines an opening through which the first rear frame mount extends; the first rear frame mount includes an upward opening notch, the upward opening notch receiving a portion of the rear wall forming part of the opening through which the rear frame mount extends; the front rail includes a front wall, the front wall defines an opening through which the first front frame mount extends; and the first front frame mount includes an upward opening notch, the upward opening notch receiving a portion of the front wall forming part of the opening through which the front frame mount extends.

22. The adjustable rack system of claim 20, wherein: the first rear frame mount includes an upward opening notch, the upward opening notch receiving a portion of the rear rail, engagement provided by the notch and the portion of the rear rail preventing movement of the first rear frame mount relative to the rear rail along the adjustment axis; and the first front frame mount includes an upward opening notch, the upward opening notch receiving a portion of the front rail, engagement provided by the notch and the portion of the front rail preventing movement of the first front frame mount relative to the front rail along the adjustment axis.

23. An adjustable rack system for mounting to a frame assembly of a shelving unit, the adjustable rack system comprising: an adjustable frame having: a front rail; a rear rail being spaced apart from the front rail; at least one longitudinally adjustable rail interconnecting the front and rear rails, the at least one longitudinally adjustable rail being longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis; and an adjustable rack removably attached to the adjustable frame, the adjustable rack defines a top surface, the adjustable rack having: a first panel defining a rear edge of the adjustable rack; and a second panel defining a front edge of the adjustable rack, the second panel being adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack, the top surface is defined in part by each of the first and the second panel; wherein the first and second panels have a sliding interlock therebetween, when coupled, the sliding interlock permits the first panel to slide relative to the second panel along the adjustment axis but prevents the first and second panels from being disconnected from one another in a direction generally orthogonal to the top surface; wherein the sliding interlock includes: at least one clip provided on one of the first and second panels; a slide region of the other one of the first and second panels that is slidably engaged by the at least one clip, the slide region being generally parallel to the adjustment axis wherein the slide region includes a notch sized to receive the at least one clip therethrough generally orthogonal to the top surface for assembling the sliding interlock.

24. An adjustable rack system for mounting to a frame assembly of a shelving unit, the adjustable rack system comprising: an adjustable frame having: a front rail; a rear rail being spaced apart from the front rail; at least one longitudinally adjustable rail interconnecting the front and rear rails, the at least one longitudinally adjustable rail being longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis; and an adjustable rack removably attached to the adjustable frame, the adjustable rack defines a top surface, the adjustable rack having: a first panel defining a rear edge of the adjustable rack; and a second panel defining a front edge of the adjustable rack, the second panel being adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack, the top surface is defined in part by each of the first and the second panel; wherein the first and second panels have a sliding interlock therebetween, when coupled, the sliding interlock permits the first panel to slide relative

to the second panel along the adjustment axis but prevents the first and second panels from being disconnected from one another in a direction generally orthogonal to the top surface; wherein the sliding interlock includes: first and second clips provided by the first panel, each of the first and second clips including a leg extending downward from a first main panel section of the first panel and a foot portion extending laterally from the leg, the foot portion and a bottom of the first main panel section forming a channel therebetween; first and second slide regions provided by a second main panel section of the second panel, the first slide region received in the channel formed by the first clip and the second slide region received in the channel formed by the second clip; wherein the second main panel includes a first notch adjacent the first slide region and a second notch adjacent the second slide region, the first and second notches sized and configured such that the foot of the first clip can pass through the first notch to align the first slide region with the channel of the first clip and the foot of the second clip can pass through the second notch to align the second slide region with the channel of the second clip.

25. The adjustable rack system of claim 24, wherein: the first notch is formed between opposed ends of the first slide region such that a first portion of the first slide region is on a first side of the first notch and a second portion of the first slide region is on a second side of the first notch; and the second notch is formed between opposed ends of the second slide region such that a first portion of the second slide region is on a first side of the second notch and a second portion of the second slide region is on a second side of the first notch.

26. An adjustable rack system for mounting to a frame assembly of a shelving unit, the adjustable rack system comprising: an adjustable frame having: a front rail; a rear rail being spaced apart from the front rail; at least one longitudinally adjustable rail interconnecting the front and rear rails, the at least one longitudinally adjustable rail being longitudinally adjustable in length to adjust the spacing between the front and rear rails along an adjustment axis; and an adjustable rack removably attached to the adjustable frame, the adjustable rack defines a top surface, the adjustable rack having: a first panel defining a rear edge of the adjustable rack; and a second panel defining a front edge of the adjustable rack, the second panel being adjustably positionable relative to the first panel to adjust a spacing between the front and rear edge of the adjustable rack, the top surface is defined in part by each of the first and the second panel; wherein: the adjustable frame includes a first rear frame mount, the first rear frame mount includes an exposed portion that extends rearward of a rear side of the rear rail, the exposed portion includes a downward opening notch; and the adjustable frame includes a first front frame mount, the first front frame mount includes an exposed portion that extends forward of a front side of the front rail, the exposed portion includes a downward opening notch; wherein: the first rear frame mount is removably attached to the rear rail, the first rear frame mount includes a first leg and a second leg extending from the first leg forming an L-shape, the first leg being longer than the second leg, the downward opening notch being formed in the exposed portion of the first leg, the first leg having a second portion located forward of the rear side of the rear rail, the second portion, at least in part, securing the first rear frame mount to the rear rail; and the first front frame mount is removably attached to the front rail, the first front frame mount includes a first leg and a second leg extending from the first leg forming an L-shape, the first leg being longer than the second leg, the downward opening notch being formed in the exposed portion of the first leg, the first leg having a second portion located rearward of the front side of the front rail, the second portion, at least in part, securing the first front frame mount to the front rail.

27. The adjustable rack system of claim 26, wherein: the rear rail includes a top wall, a bottom wall vertically spaced from the top wall, and a rear wall below the top wall; the rear wall defines the rear side of the rear rail, the rear wall defines an opening through which the first leg of the first rear frame mount extends; the bottom wall defines an opening through which the second leg of the first rear frame mount extends; the front rail includes a top wall, a bottom wall vertically spaced from the top wall, and a front wall below the top wall; the front wall defines the front side of the front

rail, the front wall defines an opening through which the first leg of the first front frame mount extends; the bottom wall defines an opening through which the second leg of the first front frame mount extends.

28. The adjustable rack system of claim 27, wherein: the rear rail includes a yoke defining a cavity receiving the second portion of the first leg of the first rear frame mount, the yoke being positioned between the rear wall and the opening in the bottom wall of the rear rail; the opening in the bottom wall of the rear rail is elongated in a lateral direction and sized to permit the first rear frame mount to be rotated from an orientation wherein the second leg is generally horizontal to an orientation wherein the second leg of the first rear frame mount extends vertically downward to insert the second leg of the first rear frame mount into the opening in the bottom wall, insertion of the second leg of the first rear frame mount into the opening in the bottom wall limits motion of the first rear frame mount parallel to the adjustment axis; the front rail includes a yoke defining a cavity receiving the second portion of the first leg of the first front frame mount, the yoke being positioned between the front wall and the opening in the bottom wall of the front rail; and the opening in the bottom wall of the front rail is elongated in a lateral direction and sized to permit the first front frame mount to be rotated from an orientation wherein the second leg of the first front frame mount is generally horizontal to an orientation wherein the second leg of the first front frame mount extends vertically downward to insert the second leg of the first front frame mount into the opening in the bottom wall, insertion of the second leg of the first front frame mount into the opening in the bottom wall limits motion of the first rear frame mount parallel to the adjustment axis.

29. A frame mount for mounting a rack frame to a shelving unit frame, the frame mount comprising: a first leg and a second leg extending from the first leg in a first direction forming an L-shape, the first leg being longer than the second leg, a first notch being formed in first leg, the notch opening in the first direction; a second notch, the second notch opening in a second direction being opposite the first direction such that the first and second notches are on opposite sides of the first leg; wherein: the second notch is located axially along the first leg between the first notch and the second leg; the first and second legs are formed from a continuous piece of material; and the continuous piece of material is a bent wire.
